STAT6011

COMPUTATIONAL STATISTICS AND BAYESIAN LEARNING

Lecture 3

Lindley's Paradox, Bayes Factor and Hierarchical Model

Bayesian Methods

- **Bayesian methods adhere to the likelihood principle: All that we know about the data/sample is contained in the likelihood function.**
- **If the likelihood functions under two different sampling plans/distributions are proportional with respect to the parameter of interest θ, statistical inferences on θ should be identical based on these two sampling distributions.**

Example

- **We consider an experiment in which a coin was tossed 12 times, with 9 heads and 3 tails observed (Lindley and Phillips, 1976).**
- **Let θ be the probability of observing a head for a toss of the coin, and we are interested in testing the hypotheses,**

$$H_0\text{: } \theta = 0.5 \quad \text{versus} \quad H_1\text{: } \theta > 0.5.$$

  There is no further information on the sampling plan.

- **Based on the observed data, there may be two choices for the likelihood function.**

- **First, let Y denote the number of heads after a fixed number of n tosses; that is, $Y \sim$ Bin(n, θ).**
- **Under the binomial distribution with $n = 12$ tosses and $y = 9$ heads observed, the likelihood function is given by**

$$L_1(\theta|y) = \binom{n}{y}\theta^y(1-\theta)^{n-y} = \binom{12}{9}\theta^9(1-\theta)^3.$$

- **Based on the binomial likelihood, the p-value is**

$$p_1\text{-value} = \Pr(y \geq 9|H_0) = \sum_{y=9}^{12}\binom{12}{y}0.5^{12} \approx 0.073.$$

- **Second, let Y be the number of heads for the tosses of the coin until the third tail $(r = 3)$ is observed; that is, $Y \sim$ Neg-Bin(r, θ).**
- **Under the negative binomial distribution, the likelihood function is given by**

$$L_2(\theta|y) = \binom{y+r-1}{y}\theta^y(1-\theta)^r = \binom{11}{9}\theta^9(1-\theta)^3.$$

- **Under the negative binomial distribution,**

$$p_2\text{-value} = \Pr(y \geq 9|H_0) = \sum_{y=9}^{\infty}\binom{y+2}{y}0.5^{3+y} \approx 0.033.$$

- **If we set the significance level at $\alpha = 0.05$, the frequentist hypothesis test yields conflicting results: The null hypothesis is accepted under the binomial distribution, but it is rejected under the negative binomial distribution.**
- **Clearly, $L_1(\theta|y) \propto L_2(\theta|y)$ except for a normalizing constant, and the posterior distributions of θ under these two sampling distributions are identical in the Bayesian framework.**
- **However, frequentist inferences about θ are very different, which depends on the sampling distribution.**

Coin Toss Example

- **Among $n = 12$ coin tosses, we observe $y = 9$ heads. Let θ denote the probability of landing on a head in a toss.**
- **We start with a two-sided hypothesis test,**

$$H_0 : \theta = 0.5 \quad \text{versus} \quad H_1 : \theta \neq 0.5.$$

- **We assign an equal prior probability to the null and alternative hypotheses, i.e., $P(H_0) = P(H_1) = 0.5$.**

- **Under the null hypothesis, the probability of observing y heads among n coin tosses is**

$$P(y|H_0) = P(y|\theta = 0.5) = \binom{n}{y}\theta^y(1-\theta)^{n-y} = \binom{n}{y}0.5^n.$$

- **Under the alternative hypothesis, we assume that θ follows a Beta(α, β) distribution, and thus**

$$\begin{aligned} P(y|H_1) &= \int_0^1 \binom{n}{y}\theta^y(1-\theta)^{n-y} \times \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\theta^{\alpha-1}(1-\theta)^{\beta-1}\mathbf{d}\theta \\ &= \binom{n}{y}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\frac{\Gamma(\alpha+y)\Gamma(\beta+n-y)}{\Gamma(\alpha+\beta+n)}. \end{aligned}$$

- **Hence, the posterior probability of H_0 can be computed using Bayes's theorem,**

$$\begin{aligned} P(H_0|y) &= \frac{P(y|H_0)P(H_0)}{P(y|H_0)P(H_0)+P(y|H_1)P(H_1)} \\ &= \frac{\theta^y(1-\theta)^{n-y}}{\theta^y(1-\theta)^{n-y}+\frac{\Gamma(\alpha+y)\Gamma(\beta+n-y)}{\Gamma(\alpha+\beta+n)}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}}. \end{aligned}$$

- **If we assume a uniform prior ($\alpha=\beta=1$) for θ under H_1,** $P(H_0|y)=0.411$**; and if we assume a beta prior with** $\alpha=\beta=0.1$**,** $P(H_0|y)=0.741$**.**

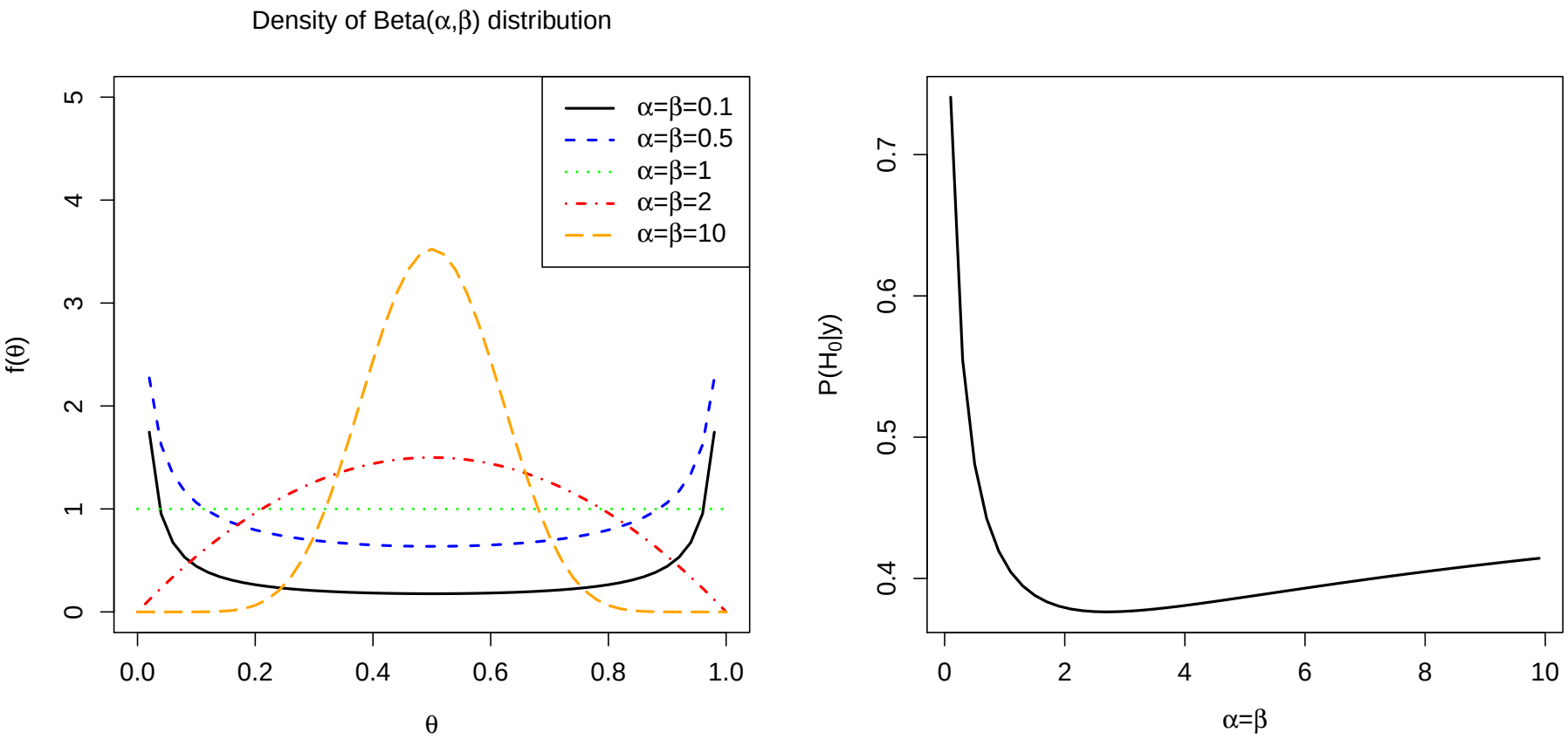


Figure 1: The posterior probability of H_0 under different beta prior distributions Beta(α, β).

$P(H_0|y)$

- **Figure 1 shows different beta prior distributions Beta(α, β) for $\alpha = \beta$ and the pattern of the posterior probability of H_0.**
- **As $P(y|H_0)$ is fixed, we focus on $P(y|H_1)$. The integration in $P(y|H_1)$ is a combination of the data likelihood and a beta prior, which is largest when the beta prior is most commensurate with the likelihood.**
- **When the beta prior is either highly noninformative (e.g., $\alpha = \beta = 0.1$) or informatively centered at 0.5 (e.g., $\alpha = \beta = 10$), both cases are not commensurate with the data likelihood.**

- **When the beta prior changes from noninformative to informative, there is a sweet spot between 0.1 and 10 where it is most commensurate with the data likelihood and gives the largest value of $P(y|H_1)$ and hence the smallest $P(H_0|y)$.**

One-sided Hypothesis Test

- **Suppose that we instead conduct a one-sided hypothesis test,**

$$H_0 : \theta = 0.5 \quad \text{versus} \quad H_1 : \theta > 0.5.$$

- **Under the alternative hypothesis, if we assume a symmetric beta prior distribution** $(\alpha = \beta)$ **with a truncated support of** $(0.5, 1)$**, and then**

$$P(y|H_1) = \int_{0.5}^{1} \binom{n}{y} \theta^y (1-\theta)^{n-y} \times 2\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1} \mathbf{d}\theta.$$

- **If we assume that θ follows a Unif$(0.5, 1)$ distribution under H_1, then**

$$P(y|H_1) = \int_{0.5}^{1} \binom{n}{y} \theta^y (1-\theta)^{n-y} \times 2\mathbf{d}\theta,$$

  which does not have an analytical form, but can be obtained via numerical integration.

- **Under a prior distribution $\theta \sim \text{Unif}(0.5, 1)$, $P(H_0|y) = 0.268$; and under a prior distribution $\theta \sim \text{Beta}(0.1, 0.1)$ for θ, $P(H_0|y) = 0.596$.**

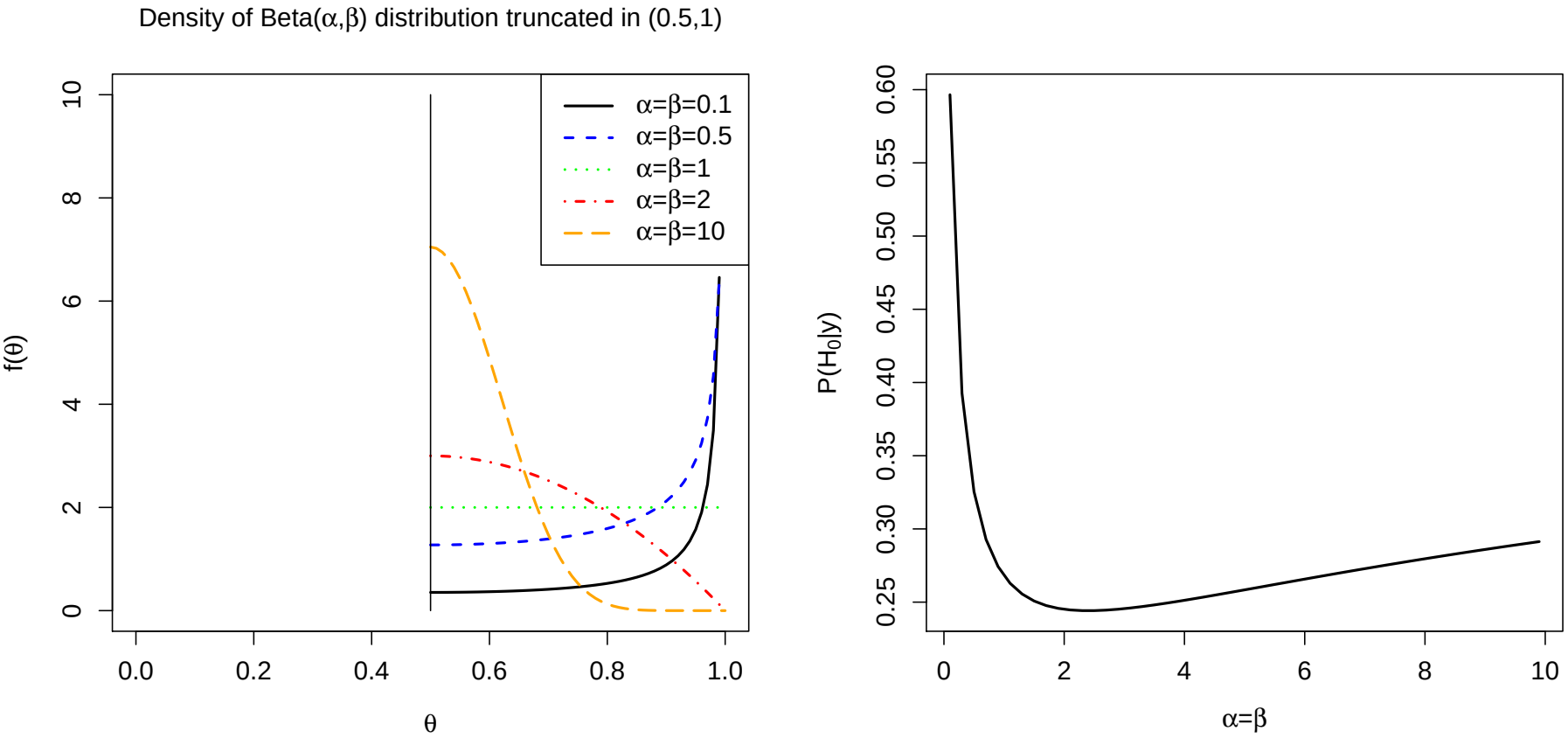


Figure 2: The posterior probability of H_0 under different truncated beta prior distributions Beta(α, β) with a support of $(0.5, 1)$.

Lindley's Paradox

- **In the hypothesis testing framework, it may happen that the Bayesian and frequentist approaches produce opposite conclusions for certain choices of the prior distribution.**
- **To illustrate Lindley's Paradox (Lindley, 1957), we start with a simple example. Suppose that 28,298 boys and 27,801 girls were born in Hong Kong last year, then the observed proportion of male births was** $y = 28298/56099 \approx 0.504$.
- **Let** θ **denote the true proportion of male births, and we are interested in testing**

$$H_0 : \theta = 0.5 \quad \text{versus} \quad H_1 : \theta \neq 0.5.$$

- **The number of male births follows a binomial distribution with mean $n\theta$ and variance $n\theta(1-\theta)$, where $n = 56,099$ is the total number of births.**
- **As the sample size n is large and the observed male proportion y is not close to 0 or 1, we can use a normal approximation to simplify the computation, so $Y \sim N(\theta, \sigma^2)$ where $\sigma^2 = \theta(1-\theta)/n$.**

Frequentist Approach

- **The frequentist approach calculates the p-value as the upper tail probability of as or more extreme than the observed data under the null distribution,**

$$\Pr(Y \geq y|H_0) = \int_{0.504}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ -\frac{(x-0.5)^2}{2\sigma^2} \right\} \mathbf{d}x \approx 0.018.$$

- **As the hypothesis test is two-sided, the final p-value is $2 \times 0.018 = 0.036$. At the typical significance level of 5%, we clearly reject H_0.**

Bayesian Approach

- **On the other hand, if we proceed with a Bayesian approach, we first specify a prior distribution on H_0 and H_1.**
- **Without any preference, we give an equal probability, i.e., $P(H_0) = P(H_1) = 0.5$. Under H_0, θ has a point mass at 0.5.**
- **Under H_1, θ is not equal to 0.5 and, to be fair, we assign a uniform prior distribution to θ on $[0, 1]$.**

- **The posterior probability of H_0 is**

$$\begin{aligned} P(H_0|y) &= \frac{P(y|H_0)P(H_0)}{P(y|H_0)P(H_0)+P(y|H_1)P(H_1)} \\ &= \frac{\exp\left\{-\frac{(y-0.5)^2}{2\sigma^2}\right\}}{\exp\left\{-\frac{(y-0.5)^2}{2\sigma^2}\right\}+\int_0^1 \exp\left\{-\frac{(y-\theta)^2}{2\sigma^2}\right\}\mathbf{d}\theta} \\ &\approx 0.954, \end{aligned}$$

  which strongly supports H_0.

- **Such conflict between Bayesian and frequentist hypothesis testing approaches may happen when the prior distribution is a mixture of a sharp peak at H_0 and no sharp features anywhere else.**

Credible Interval

- In the frequentist paradigm, the parameter is fixed, but the confidence interval is random.
- The usual 95% confidence interval has an interpretation that by replicating the same experiment for a large number of times, the probability that these intervals contain the true parameter is 0.95.
- By contrast, the Bayesian credible interval given the data is fixed, while the parameter is considered to be random.
- Hence, the probability that the parameter is covered by (or falls inside) the 95% credible interval is 0.95.

Credible Interval

- **Let $f(\theta|\mathbf{y})$ be the posterior distribution of θ, and let Θ be the support of θ.**
- **For a subset of Θ, say $\mathcal{A}$, the posterior probability of the credible set $\mathcal{A}$ is**

$$\Pr(\theta \in \mathcal{A}|\mathbf{y}) = \int_{\mathcal{A}} f(\theta|\mathbf{y})\, \mathbf{d}\theta.$$

- **Let $F(\theta|\mathbf{y})$ denote the posterior cumulative distribution function of θ, and let $F^{-1}(\alpha|\mathbf{y})$ denote the αth quantile of θ, where $F^{-1}(\cdot|\mathbf{y})$ is the inverse function. $100(1-\alpha)\%$ credible interval:** $\left[F^{-1}(\alpha/2|\mathbf{y}),\ F^{-1}(1-\alpha/2|\mathbf{y})\right]$.

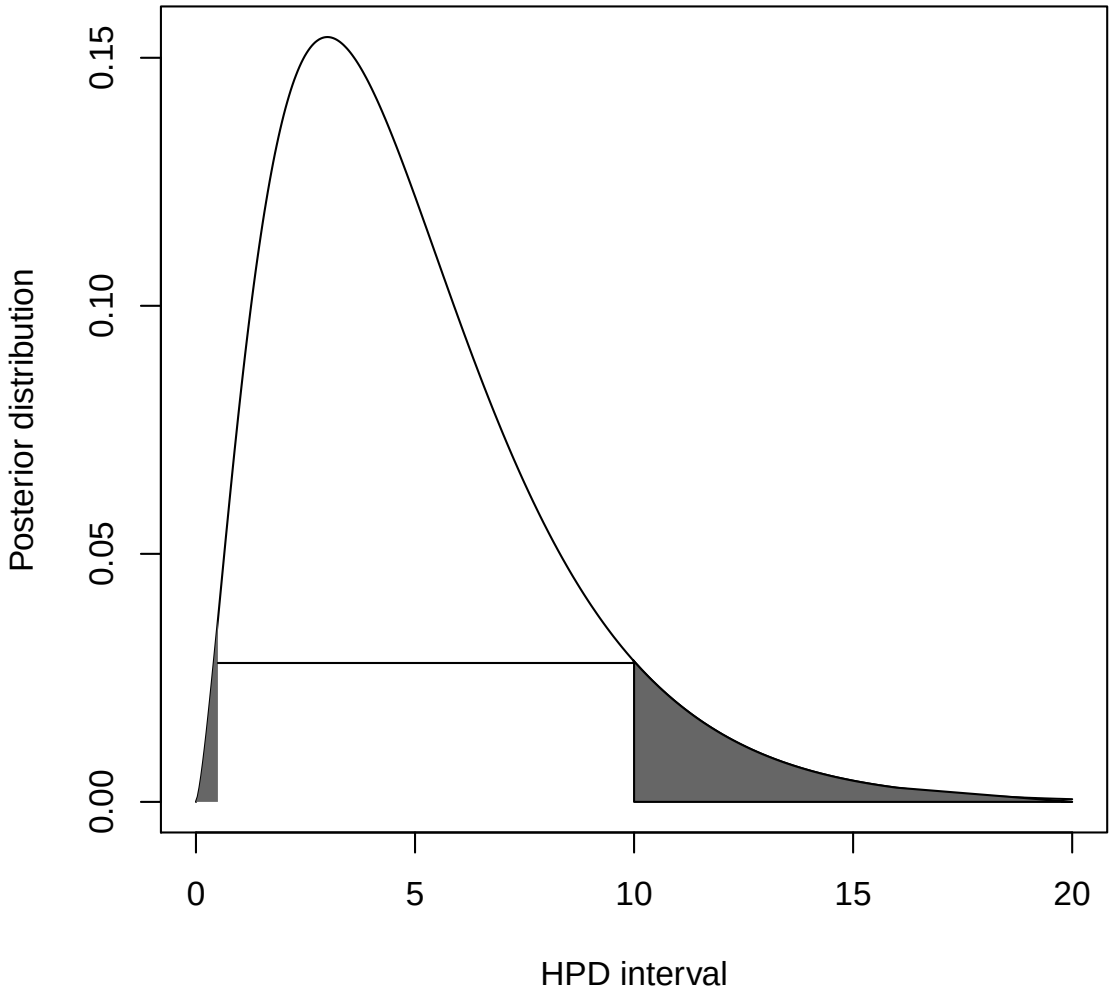


Figure 3: Illustration of the highest posterior density (HPD) interval.

HPD Interval

- **A more meaningful Bayesian interval estimate is called the highest posterior density (HPD) interval.**
- **As shown in Figure 3, if $f(\theta|\mathbf{y})$ is unimodal, the $100(1-\alpha)\%$ HPD interval for θ is given by**

$$\mathcal{A}(\pi_\alpha) = \{\theta\text{: } f(\theta|\mathbf{y}) \geq \pi_\alpha\},$$

where π_α is the largest constant such that

$$\int_{\mathcal{A}(\pi_\alpha)} f(\theta|\mathbf{y})\mathbf{d}\theta = 1-\alpha.$$

- **If $f(\theta|\mathbf{y})$ is symmetric and unimodal, the credible interval and the HPD interval coincide.**

Marginal Likelihood

- As the model becomes more complex with more parameters involved, the likelihood function always increases, due to the fact that more complex models always fit the data better than simpler models.
- However, this may lead to model overfitting, which is particularly dangerous for prediction.
- In contrast, the marginal likelihood has a natural built-in penalty for more parameters.

- **The marginal likelihood for model M is obtained by integrating out the model parameter in the likelihood function with respect to the prior distribution,**

$$P(D|M) = \int L(D|\theta, M) f(\theta|M) \, \mathbf{d}\theta.$$

- **Intuitively, as the complexity of the model increases, the prior is more spread out across all the models.**
- **By integrating the likelihood with respect to the thinly spreading prior, the marginal likelihood may start to decrease when too little prior mass is placed on the good models.**

- **Under model M, the posterior distribution of θ is**

$$P(\theta|D, M) = \frac{P(D|\theta, M)P(\theta|M)}{P(D|M)},$$

which leads to the marginal likelihood in the form of

$$P(D|M) = \frac{P(D|\theta, M)P(\theta|M)}{P(\theta|D, M)}. \tag{1}$$

- **Taking the logarithm on both sides of (2), we have**

$$\log P(D|M) = \log P(D|\theta, M) - \{\log P(\theta|D, M) - \log P(\theta|M)\},$$

where the log-likelihood $\log P(D|\theta, M)$ always increases with the number of parameters, but $\log P(\theta|D, M) - \log P(\theta|M)$ can be viewed as a penalty for model complexity.

- **This is in a similar spirit to the Akaike (1973) information criterion (AIC) and the Bayesian information criterion (BIC), which also penalize the number of model parameters,**

$$\begin{aligned} \mathrm{AIC} &= -2\log L + 2\nu, \\ \mathrm{BIC} &= -2\log L + \nu \log n, \end{aligned}$$

  where $\log L$ is the log-likelihood and ν is the number of model parameters.

- **The smaller the AIC or BIC, the better the model fit.**

Bayes Factor

- **Given the observed data D, we consider two candidate models, M_0 and M_1, characterized by parameters θ_0 and θ_1, respectively.**
- **Let $L(D|\theta_k, M_k)$ denote the likelihood function under model M_k for $k = 0, 1$.**
- **We specify the prior distributions for θ_0 and θ_1 as $f(\theta_0|M_0)$ and $f(\theta_1|M_1)$, and the prior model probabilities for M_0 and M_1 as $P(M_0)$ and $P(M_1)$, respectively.**

- **The marginal likelihood for model M_k is obtained by integrating out the model parameter in the likelihood function with respect to the prior distribution,**

$$P(D|M_k) = \int L(D|\theta_k, M_k) f(\theta_k|M_k)\, \mathbf{d}\theta_k. \tag{2}$$

- **The posterior model probability for M_k is given by**

$$P(M_k|D) = \frac{P(D|M_k)P(M_k)}{P(D|M_0)P(M_0) + P(D|M_1)P(M_1)}.$$

Bayes Factor

- **Noting that $P(M_1)/P(M_0)$ is the prior odds in favor of model M_1 over M_0, the posterior odds is given by**

$$\frac{P(M_1|D)}{P(M_0|D)} = \frac{P(M_1)}{P(M_0)} \times \frac{P(D|M_1)}{P(D|M_0)}.$$

- **The Bayes factor (BF) is defined as the ratio of the posterior odds to the prior odds in favor of model M_1 over M_0,**

$$\mathrm{BF}_{1,0} = \frac{P(M_1|D)/P(M_0|D)}{P(M_1)/P(M_0)} = \frac{P(D|M_1)}{P(D|M_0)}.$$

Example of Bayes Factor

- **Suppose that we observe $y = 13$ responders among a total of $n = 16$ treated patients.**
- **Let θ denote the response rate, and we are interested in testing**

$$H_0 : \theta < 0.6 \quad \text{versus} \quad H_1 : \theta \geq 0.6.$$

- **If we assume a uniform prior distribution for θ, i.e., $\theta \sim \text{Beta}(1, 1)$, then $P(H_0) = P(\theta < 0.6) = 0.6$ and $P(H_1) = 0.4$.**
- **Given the binomial likelihood $y \sim \text{Bin}(n, \theta)$, the posterior distribution of θ is $\text{Beta}(y + 1, n - y + 1) = \text{Beta}(14, 4)$, and then $P(H_0|y) = P(\theta < 0.6|y) = 0.046$ and $P(H_1|y) = 0.954$.**

- **This leads to the Bayes factor in favor of H_1 over H_0 as**

$$\mathrm{BF}_{1,0} = \frac{P(H_1|y)/P(H_0|y)}{P(H_1)/P(H_0)} = \frac{0.954/0.046}{0.4/0.6} = 31.1,$$

  which implies strong evidence against H_0.

- **On the other hand, we can first give an equal probability to H_0 and H_1, i.e., $P(H_0) = P(H_1) = 0.5$, and then assume a uniform prior distribution for θ under H_0 and H_1 respectively, i.e., $p(\theta|H_0) = 1/0.6$ and $p(\theta|H_1) = 1/0.4$.**

- **The Bayes factor in favor of H_1 over H_0 is**

$$\begin{aligned}
\mathrm{BF}_{1,0} &= \frac{P(H_1|y)/P(H_0|y)}{P(H_1)/P(H_0)} = \frac{P(y|H_1)}{P(y|H_0)} \\
&= \frac{\int_{0.6}^{1} p(y|\theta, H_1)p(\theta|H_1)\mathbf{d}\theta}{\int_{0}^{0.6} p(y|\theta, H_0)p(\theta|H_0)\mathbf{d}\theta} \\
&= \frac{\int_{0.6}^{1} \theta^{13}(1-\theta)^{3}\mathbf{d}\theta / \int_{0}^{0.6} \theta^{13}(1-\theta)^{3}\mathbf{d}\theta}{0.4/0.6} = 31.1.
\end{aligned}$$

- **Therefore, both ways lead to the same conclusion.**

Posterior Odds

- **Similarly in the Bayesian hypothesis testing, let $P(H_0)$ and $P(H_1)$ denote the prior probabilities assigned to the null and alternative hypotheses, respectively.**
- **The posterior odds of favoring H_1 over H_0 is**

$$\frac{P(H_1|D)}{P(H_0|D)} = \frac{P(H_1)}{P(H_0)} \times \frac{P(D|H_1)}{P(D|H_0)} = \frac{P(H_1)}{P(H_0)} \times \mathrm{BF}_{1,0}.$$

  Therefore, the Bayes factor $\mathrm{BF}_{1,0}$ can be interpreted as the weight or strength of evidence contained in the data to support H_1 against H_0 (Jeffreys, 1961; Kass and Raftery, 1995).
- **In the case of testing two simple hypotheses, H_0: $\theta = \theta_0$ versus H_1: $\theta = \theta_1$, the Bayes factor reduces to the likelihood ratio.**

Bayesian Hypothesis Testing

- **Quantify the strength of the data information in favor of H_1 against H_0.**
- **If $0 < \log_{10}(\mathrm{BF}_{1,0}) \leq 1/2$, the evidence in the data favoring H_1 over H_0 is not worth more than a bare mention.**
- **If $1/2 < \log_{10}(\mathrm{BF}_{1,0}) \leq 1$, the data contain substantial evidence against H_0.**
- **If $1 < \log_{10}(\mathrm{BF}_{1,0}) \leq 2$, the data contain strong such evidence.**
- **If $\log_{10}(\mathrm{BF}_{1,0}) > 2$, the evidence in favor of H_1 against H_0 appears to be decisive.**

Bayesian Model Averaging

- **Suppose that there are K candidate models, $M_1, \ldots, M_K$, to fit the observed data D. Let $P(M_k)$ be the prior probability that M_k is the true model, $k = 1, \ldots, K$.**
- **If there is no preference *a priori* to any single model, an equal probability mass is assigned to each model by simply setting $P(M_k) = 1/K$, which represents a discrete uniform prior distribution.**
- **Let θ_k be the parameter associated with model M_k, and let $f(\theta_k|M_k)$ be the prior distribution of θ_k.**

BMA

- **For $k = 1, \ldots, K$, the posterior model probability for M_k is given by**

$$P(M_k|D) = \frac{P(D|M_k)P(M_k)}{\sum_{j=1}^{K} P(D|M_j)P(M_j)},$$

where $P(D|M_k)$ is the marginal likelihood under model M_k.

- **The Bayes factor $\mathrm{BF}_{k,1}$ for model M_k against the reference model M_1 is the ratio of the marginal likelihoods,**

$$\mathrm{BF}_{k,1} = \frac{P(M_k|D)/P(M_1|D)}{P(M_k)/P(M_1)} = \frac{P(D|M_k)}{P(D|M_1)}.$$

- **We can construct such a Bayes factor for each of the K models versus model M_1, $\mathrm{BF}_{1,1}, \ldots, \mathrm{BF}_{K,1}$.**

- **Then, the posterior model probability of M_k can be naturally linked to the Bayes factors,**

$$P(M_k|D) = \frac{\mathrm{BF}_{k,1}P(M_k)}{\sum_{j=1}^{K}\mathrm{BF}_{j,1}P(M_j)} = \frac{\mathrm{BF}_{k,1}}{\sum_{j=1}^{K}\mathrm{BF}_{j,1}},$$

  if all the models are equally probable *a priori*.

- **The Bayesian model averaging (BMA) procedure provides a coherent mechanism to account for the uncertainty associated with each candidate model (Raftery, Madigan, and Hoeting, 1997).**

- **For the model parameter θ, the BMA estimator is given by**

$$\bar{\theta} = \sum_{k=1}^{K} \hat{\theta}_k P(M_k|D),$$

where $\hat{\theta}_k$ is the posterior mean of θ under model M_k,

$$\hat{\theta}_k = \int \theta_k \frac{L(D|\theta_k, M_k) f(\theta_k|M_k)}{\int L(D|\theta_k, M_k) f(\theta_k|M_k)\, \mathbf{d}\theta_k} \mathbf{d}\theta_k.$$

- **By assigning the posterior mean $\hat{\theta}_k$ a weight of $P(M_k|D)$, BMA automatically lean toward the best fitting model, and thus $\bar{\theta}$ will be close to the best parameter estimate (Madigan and Raftery, 1994; Hoeting et al., 1999).**

Bayesian Hierarchical Model

- **Bayesian hierarchical modeling often contains several layers of hierarchies, which is typically used to model dependent data, borrow information or strength across different subgroups, or pool separate studies together for joint inference.**
- **For illustration, we consider a longitudinal study, in which each subject is repeatedly measured over time.**

- **For $i = 1, \ldots, n$ and $k = 1, \ldots, K$, the linear random effects model, may be used to account for the intra-patient correlation,**

$$y_{ik} = \boldsymbol{\beta}^\top \mathbf{Z}_{ik} + \mathbf{b}_i^\top \mathbf{X}_{ik} + \epsilon_{ik}, \tag{3}$$

where $\boldsymbol{\beta}$ is the p-dimensional fixed effects, $\mathbf{b}_i$ is the q-dimensional random effects, $\mathbf{b}_i \sim \mathrm{N}_q(\mathbf{0}, \mathbf{G})$, and the error $\epsilon_{ik} \sim \mathrm{N}(0, \sigma^2)$.

- **The Bayesian hierarchical model for (3) can be formulated as**

$$
\begin{aligned}
y_{ik}|\mathbf{b}_i, \sigma^2 &\sim \mathrm{N}(\boldsymbol{\beta}^\top \mathbf{Z}_{ik} + \mathbf{b}_i^\top \mathbf{X}_{ik}, \sigma^2), \\
\mathbf{b}_i|\mathbf{G} &\sim \mathrm{N}_q(0, \mathbf{G}), \\
\mathbf{G} &\sim \textbf{Inv-Wishart}((\eta\boldsymbol{\Omega})^{-1}, \eta), \\
\sigma^2 &\sim \mathrm{IG}(\xi, \xi),
\end{aligned}
$$

where Ω is a $q \times q$ symmetric and positive definite matrix, η is a scalar parameter, and ξ is a hyperparameter.

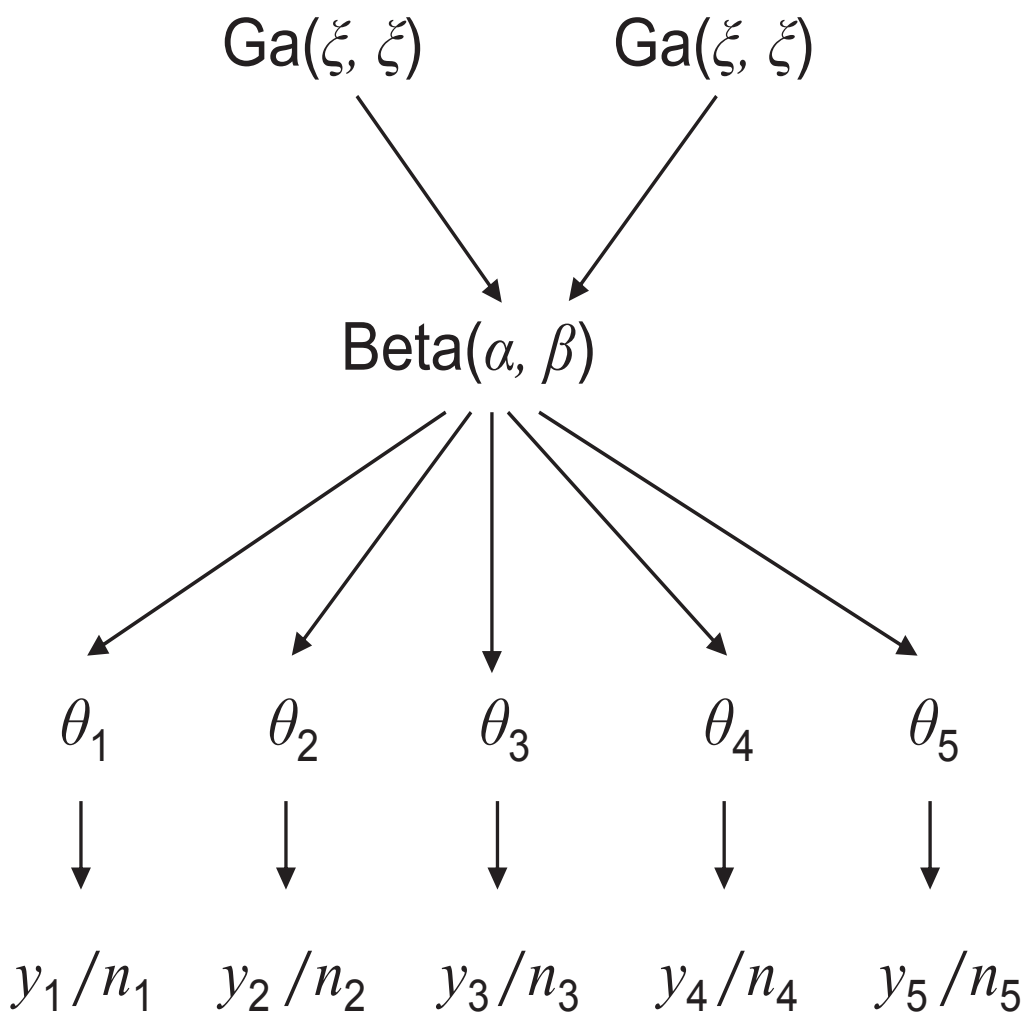
Example

- **In a multi-arm randomized trial, it is desirable to borrow information across different subgroups.**
- **Let $\theta_1, \ldots, \theta_5$ denote the response rates for the five treatments, respectively. Suppose that we observe y_k responses among n_k patients treated in arm k, $k = 1, \ldots, 5$.**

- **The Bayesian hierarchical model can be formulated as follows,**

$$\begin{aligned} y_k|\theta_k &\sim \text{Bin}(n_k, \theta_k), \\ \theta_k|\alpha, \beta &\sim \text{Beta}(\alpha, \beta), \\ \alpha &\sim \text{Ga}(\xi, \xi), \\ \beta &\sim \text{Ga}(\xi, \xi), \end{aligned}$$

where the hyperparameter ξ in the gamma distribution may take a small value (e.g., $\xi = 0.001$) to induce noninformative prior distributions.



Hyperprior

Hierarchical prior

Response rate

Observed data

Example

- **Let y** $= (y_1, \ldots, y_5)$ **denote the observed data.**
- **The joint posterior distribution is given by**

$$f(\theta_1, \ldots, \theta_5, \alpha, \beta|\mathbf{y}) \propto \left\{ \prod_{k=1}^{5} L(\theta_k|y_k) f(\theta_k|\alpha, \beta) \right\} f(\alpha) f(\beta),$$

where the binomial likelihood function is

$$L(\theta_k|y_k) = \binom{n_k}{y_k} \theta_k^{y_k} (1-\theta_k)^{n_k - y_k},$$

$f(\theta_k|\alpha, \beta)$ **denotes the beta density function, and** $f(\alpha)$ **and** $f(\beta)$ **denote the gamma prior distributions.**

- **The full conditional distribution of θ_k is**

$$\theta_k|\alpha, \beta, \mathbf{y} \sim \text{Beta}(\alpha + y_k, \beta + n_k - y_k),$$

while those of α and β do not have closed forms.

Remarks

- **If the prior distributions on α and β are removed, i.e., α and β are fixed constants,**

$$y_k|\theta_k \sim \text{Bin}(n_k, \theta_k),$$
$$\theta_k|\alpha, \beta \sim \text{Beta}(\alpha, \beta),$$

the data in the five arms are treated completely independent (without any borrowing of information across arms).

- **On the other hand, if we let all the θ_k's to be equal, the data in five groups are essentially pooled together without considering potential heterogeneity; that is,**

$$y|\theta \sim \text{Bin}(n, \theta),$$
$$\theta|\alpha, \beta \sim \text{Beta}(\alpha, \beta),$$

  where $\theta_1 = \cdots = \theta_5 = \theta$, $y = \sum_{k=1}^{5} y_k$ and $n = \sum_{k=1}^{5} n_k$.

- **The hierarchical model takes a compromising approach between the two extreme cases by adapting the degree of information borrowing to the data across five groups.**

Decision Theory

- **Bayesian decision theory provides a statistical foundation for quantifying the trade-offs between decisions and the associated rewards/costs (Berger, 1985).**
- **Let r denote a reward, and let $\mathcal{U}(r)$ denote a utility function.**
- **The expected utility under a probability distribution P is written as $\mathbf{E}_P\{\mathcal{U}(r)\}$.**
- **The probability distribution P_2 is preferred to P_1 if and only if $\mathbf{E}_{P_1}\{\mathcal{U}(r)\} < \mathbf{E}_{P_2}\{\mathcal{U}(r)\}$.**
- **Let the parameter θ denote the state of nature, which has an influence on the decision process.**

- **The loss function is the opposite of an utility function, which is associated with the cost for a decision.**
- **If an action a is taken, a loss $\mathcal{L}(\theta, a)$ may be incurred.**

Loss Functions

- **The squared-error loss function takes the form of**

$$\mathcal{L}(\theta, a) = (\theta - a)^2,$$

which has an analogy to the least-squares estimation.

- **The linear loss function specifies**

$$\mathcal{L}(\theta, a) = \begin{cases} c_0(\theta - a), & \text{if } \theta - a \geq 0, \\ c_1(a - \theta), & \text{if } \theta - a < 0, \end{cases}$$

which reduces to the absolute-error loss function $\mathcal{L}(\theta, a) = |\theta - a|$, if the constants c_0 and c_1 are the same; for example, $c_0 = c_1 = 1$.

- **The zero-one loss function is given by**

$$\mathcal{L}(\theta, a_j) = \begin{cases} 0, & \text{if } \theta \in \Theta_j, \\ 1, & \text{if } \theta \in \Theta_k \ (k \neq j). \end{cases}$$

In a two-action decision problem, if $\theta \in \Theta_0$, action a_0 is correct with no loss and action a_1 is incorrect with a loss of 1.

Posterior Bayes Action

- **Based on the likelihood $L(\theta|\mathbf{y})$ and the prior distribution $f(\theta)$, the posterior distribution of θ is given by**

$$f(\theta|\mathbf{y}) \propto L(\theta|\mathbf{y}) f(\theta).$$

- **In the posterior decision analysis, we can compute the posterior expected loss for an action a,**

$$\mathbf{E}\{\mathcal{L}(\theta, a)|\mathbf{y}\} = \int_{\Theta} \mathcal{L}(\theta, a) f(\theta|\mathbf{y}) \, \mathbf{d}\theta, \tag{4}$$

and by minimizing (4) with respect to a, we obtain a posterior Bayes action.